

David Almona

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EDUCATION

Centre College

Danville, KY

Bachelor of Science, Economics; Minor, Data Science

May 2026

- Cumulative GPA: 3.64 | Major GPA: 3.78
- Relevant Coursework: Applied Machine Learning, Statistical Modeling, Econometrics, Empirical Analysis, Data Visualization, Modern Calculus I, Managerial Finance, Programming and Problem Solving

RESEARCH EXPERIENCE

Undergraduate Research Fellow

Jun 2025 – Jul 2025

Carnegie Mellon University, Department of Statistics & Data Science

Pittsburgh, PA

- Competitively selected as 1 of 16 students for a fully funded 8-week summer research program
- Developed machine learning models to predict pressing effectiveness in professional soccer, analyzing 252,646 pressing sequences from MLS regular season games using 10Hz player tracking data provided by SkillCorner
- Collaborated with a fellow from Carnegie Mellon University and Head of Analytics at Charlotte FC
- Presented findings at the Carnegie Mellon Sports Analytics Conference (Oct 2025) and the American Soccer Insights Summit (Jan 2026)

Undergraduate Researcher

Sep 2024 – Nov 2024

Centre College

Danville, KY

- Built an interactive RShiny dashboard analyzing shot data for the Centre College Division III women's soccer team
- Presented findings in a poster session at the Carnegie Mellon Sports Analytics Conference (Nov 2024)

PROFESSIONAL EXPERIENCE

Basketball Analytics Intern

Feb 2026 – Present

Los Angeles Sparks (WNBA)

Remote

- Create statistical player comparisons and analyze individual and team performance to support roster and game strategy decisions
- Scrape and compile data from public websites to build and maintain internal analytics databases and projection models
- Contribute to game-by-game, seasonal, and player evaluation reports, including league-wide trend research

Business Program Development Fellow

Sep 2025 – Dec 2025

Sustainable Business Network (SBN) of Massachusetts

Cambridge, MA

- Developed the Technical Assistance Pilot Program from scratch to connect local food micro-entrepreneurs with specialized consultants
- Designed multi-step intake process including dual intake forms (20+ questions each), interview framework, database management system, and memorandum of understanding template
- Facilitated consultee interviews to identify foundational business gaps and matched organizations with consultants based on readiness, expertise alignment, and availability

Data Analytics: Qualitative & Quantitative Insights Extern

Mar 2025 – Jun 2025

Beats by Dr. Dre

Remote

- Analyzed 4,628 consumer survey responses and scraped product reviews using Python (Pandas, NumPy, TextBlob) to identify market opportunities and feature preferences for a wireless speaker product launch

- Delivered strategic assessment recommending \$150–250 price point based on consumer insights, demographic segmentation, sentiment analysis, and competitive gap analysis

PUBLICATIONS & PRESENTATIONS

- Almona, D., Rayce, N., & Wicker, D. (2026, January). [*Forced Turnover: Evaluating Pressing Effectiveness in Soccer*](#). Oral presentation at the American Soccer Insights Summit.
- Almona, D., Rayce, N., & Wicker, D. (2025, October). [*Forced Turnover: Evaluating Pressing Effectiveness in Soccer*](#). Oral presentation at the Carnegie Mellon Sports Analytics Conference, Pittsburgh, PA.
- Almona, D., Rayce, N., & Wicker, D. (2025, July). [*Forced Turnover: Evaluating Pressing Effectiveness in Soccer*](#). Research report. Carnegie Mellon University.
- Almona, D. (2024, November). *Centre College Women's Soccer Analytics Dashboard*. Poster session, Carnegie Mellon Sports Analytics Conference, Pittsburgh, PA.

PROJECTS

[Chasing Imperfect Passes: Into the Life of a Receiver](#) Sep – Dec 2025
 2026 NFL Big Data Bowl | R | Partner: Sammy Davis

- Analyzed how quarterback ball placement alters receiver movement while the ball is in the air, using player tracking data from NFL games
- Developed a placement-based metric capturing receiver adjustment through trajectory deviation, acceleration, and turning behavior

[Forced Turnover: Evaluating Pressing Effectiveness in Soccer](#) Jun – Jul 2025
 CMU Sports Analytics Camp | R | Partners: Natalie Rayce (CMU), Daniel Wicker (Charlotte FC)

- Developed ML classification models on 252,646 pressing sequences using 10Hz SkillCorner tracking data from MLS regular season games
- Evaluated model performance and identified key predictive features of successful pressing in professional soccer

[Loan Default Prediction](#) Mar 2026
 Applied Machine Learning Course | Python

- Modeled loan default risk on 307,000+ Home Credit applicants, comparing 6 ML algorithms with 5-fold cross-validation
- Tuned decision threshold via F2 score to prioritize recall, reducing costly false negatives for the lender

[Predicting Pulsars & Forest Cover](#) Apr 2026
 Applied Machine Learning Course | Python

- Evaluated 5 classifiers on 48,000+ rows using scikit-learn Pipeline and 5-fold GridSearchCV
- Achieved 0.728 macro F1 on 7-class forest cover prediction; SVM (RBF) and Random Forest tied for top performance

[NBA Honors Selections, 2024–25 Season](#) May 2025
 Course Project | R | Partner: Vedant Bhong

- Predicted end-of-season NBA honors (All-NBA, All-Defensive, All-Rookie teams) using regular-season box score statistics
- Research featured in Centre College campus news coverage

[Predicting MLB Player Usage](#) Feb 2025
 Cincinnati Reds Baseball Analytics Student Hackathon | R

- Predicted total playing time (plate appearances / batters faced) for MLB players using historical performance and usage data

[National Olympic Performance: Determinants of Medal Counts](#) Dec 2024
 Econometrics Final Paper | Stata

- Built econometric models identifying macroeconomic and demographic predictors of national Olympic medal counts across countries

Centre College Women's Soccer Analytics Dashboard

Nov 2024

R, RShiny

- Built an interactive dashboard to visualize and analyze shot data for a Division III collegiate women's soccer team

HONORS & AWARDS

CSDS Conference Travel Award

Feb 2026

Texas A&M University & Collaborative for Sport Data Science (CSDS)

- Competitive funding supporting student attendance and presentation of sport data science research at leading academic conferences

CFSI Values Award

Dec 2025

College for Social Innovation

- Recognized for capacity building and lasting organizational impact during fellowship placement

Academic All-District Honor Roll

Jun 2024

College Sports Communicators (CSC)

Dean's List (×3)

Centre College | Awarded for cumulative GPA ≥ 3.60

Combs Achievement Scholarship

Aug 2024

Centre College

Colonel Scholarship

Aug 2022

Centre College

TECHNICAL SKILLS

Programming Languages: Python, R, Stata, SQL

Machine Learning & Statistics: scikit-learn, caret, XGBoost, Random Forest, SVM, logistic regression, cross-validation, NLP/sentiment analysis

Data Wrangling & Analysis: Pandas, NumPy, TextBlob, tidyverse, dplyr, data.table

Visualization & Reporting: ggplot2, matplotlib, seaborn, RShiny, Tableau, Power BI, Jupyter Notebook, Quarto

Developer Tools: Git, GitHub, Microsoft Excel, Trello

LEADERSHIP & SERVICE

Co-President, Centre Brother-to-Brother Club

Centre College

Resident Assistant

Centre College

Student-Athlete, Track and Field (100m, 200m)

Centre College

MENTORSHIP & PROFESSIONAL DEVELOPMENT

- Mentee, 2026 NFL Big Data Bowl Mentorship Program (selected from competitive applicant pool)
- Mentee, WeStutter@Work Mentorship Program
- Selected Participant, Philadelphia Eagles Virtual Development Series (Business Operations)